## Supplementary Table 2 Inclusion/exclusion criteria

| | Included | Excluded |
|---|---|---|
| **Population of interest**<br><br>Apparently healthy children and adults | • Healthy children of any age<br>• Healthy adults up to 65 years of age | • Those with illness that might impact vitamin D or calcium status or metabolism<br>• Pregnancy or lactation |
| **Outcomes**<br><br>Measures of vitamin D status | • Serum or plasma 25(OH)D levels | • Studies not reporting outcome of interest<br>• Studies not reporting baseline and endpoint concentrations |
| **Interventions**<br><br>Exposure to sunlight or artificial UVB | • For studies with natural sunlight exposure, quantification of personal exposure (personal dosimeter and/or exposure diary and ambient measurements for sunlight)<br>• Defined dose of artificial UV<br>• For studies using artificial UVB, the source must be clearly specified, with spectral information in the paper or available because a standard lamp type. | • Studies providing high-dose UVB for therapeutic purposes (i.e. dose greater than MED (minimum erythema dose)<br>• Studies not quantifying natural sunlight exposure<br>• Studies not reporting source of UVB or sufficient detail regarding the source of UVB |
| **Study designs** | • Intervention studies; including randomized and non-randomized controlled trials, non-controlled intervention studies (i.e. before and after studies) | • Observational studies including cohort, case-control, cross-sectional, ecological studies<br>• Animal studies<br>• Reviews (but use for snowball referencing). |
| **Study details** | • Studies of any duration<br>• Minimum sample size of N>2 | • Individual/ case-control studies<br>• Studies with participants taking vitamin D supplements |
| **Publication type** | • Peer-reviewed journal publications | • Conference posters and abstracts<br>• Articles for which full-text unobtainable |
| **Other**<br><br>No restrictions on dates, settings, languages (WHO will translate if needed) | • All dates, settings, languages | |